Control work 4, and the quarter review

Irrational equations and systems in one paper, then the map that links a domain to a check to an answer.

LESSON 47–48 | 2 × 40 MIN | ALGEBRA 10, NAZORAT ISHI 4 | P1 · CHAPTER 1 REVIEW

LESSON CLOCK

3 | 42' | 10' | 20' | 10' | 5'

Instructions	3 min
The paper	42 min
Answers	10 min
Rewrite	20 min
Concept map	10 min
Targets	5 min

LEARNING OBJECTIVES

- Apply every method of Quarter II in one assessment.
- Decide which technique a question wants without being told.
- Build a concept map linking domain, transformation and check.
- Set a personal target for Quarter III.

TEXTBOOK

National	pp. 181–184
Cambridge	Pure Mathematics 1, pp. 24–25
Workbook	Control paper D

01 Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Solve $\sqrt{2x+7} = x + 2$, with the conditions stated	5	L40–42
2	Solve $\sqrt{x+5} - \sqrt{x} = 1$	5	L40–42
3	Solve $x - 6\sqrt{x} + 8 = 0$ by substitution	4	L40–42
4	Solve $\sqrt{x} + \sqrt{y} = 6,\ x + y = 20$	6	L43–46
5	Solve $\dfrac{2}{x-1} \geq 1$ with a sign chart	5	L32–34
6	Find the domain of $\dfrac{\sqrt{x-2}}{x^2-25}$	5	L35–37

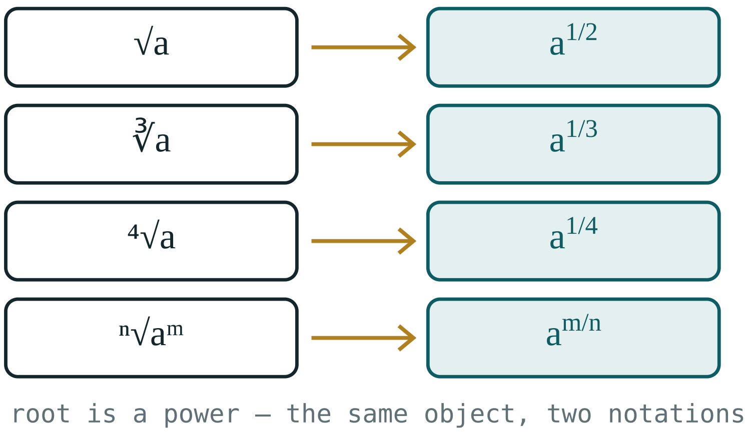
TWO MARKS ARE FOR CONDITIONS

Q1 gives one for $x + 2 \ge 0$ and one for the rejection that follows from it. A correct answer with neither scores three of five.

The concept map

Six boxes, links as sentences:

- **rational equation** → **domain** — “each denominator forbids one value”
- **domain** → **extraneous root** — “multiplying by zero is not reversible”
- **rational inequality** → **sign chart** — “never cross-multiply”
- **system** → **intersection** — “draw them on one line”
- **irrational equation** → **two conditions** — “radicand ≥ 0 and the other side ≥ 0”
- **squaring** → **check** — “sign information is destroyed, so it must be restored”



The single picture behind the whole quarter: a root never goes below the axis.

Looking forward

Quarter III opens with the exponential and logarithmic functions. Their equations are solved by exactly the technique of this quarter: state the domain, transform, check. A logarithm demands $argument > 0$ before anything else — the same first line as $\sqrt{A}$ demands $A \ge 0$.

ONE HABIT TO CARRY FORWARD

The first line of every solution states what the unknown is allowed to be. Learners who built that habit this quarter will find Quarter III half-solved before they start.

02 Worked examples

EXAMPLE 1 **Model answer, Q1: solve $\sqrt{2x+7} = x + 2$.**

1. Conditions $2x + 7 \geq 0$ and $x + 2 \geq 0$, so $x \geq -2$.
2. $2x + 7 = x^2 + 4x + 4$
3. $x^2 + 2x - 3 = 0 \Rightarrow x = 1, -3$
4. $x = -3$ fails $x \geq -2$.

ANSWER

$x = 1$

EXAMPLE 2 **Model answer, Q4: $\sqrt{x} + \sqrt{y} = 6, x + y = 20$.**

1. $u + v = 6$, $u^2 + v^2 = 20$, both ≥ 0.
2. $uv = \dfrac{36 - 20}{2} = 8$
3. $t^2 - 6t + 8 = 0 \Rightarrow t = 2, 4$
4. $x = 4, y = 16$ or the reverse.

ANSWER

$(4, 16)$ and $(16, 4)$

EXAMPLE 3 **Model answer, Q6: domain of $\dfrac{\sqrt{x-2}}{x^2-25}$.**

1. $x - 2 \geq 0 \Rightarrow x \geq 2$
2. $x^2 - 25 \neq 0 \Rightarrow x \neq \pm 5$
3. $x = -5$ is already excluded.

ANSWER

$x \geq 2,\ x \neq 5$

03 Interactive model

Work Q1 and Q4 on the board with the conditions written in a box at the top.

The quarter in ten questions

Domain of $\frac{1}{x-4} + \frac{1}{x}$:

- $x \neq 4$
- $x \neq 0$
- $x \neq 0, 4$
- $\mathbb{R}$

$\frac{x^2}{x-3} = \frac{9}{x-3}$ gives:

- ± 3
- 3
- -3
- none

$\frac{1}{x} > 2$ gives:

- $x < 0.5$
- $0 < x < 0.5$
- $x > 0.5$
- $x > 2$

$\underline{x - 4}\;\; x + 2 \;\le 0$ gives:

- $-2 \le x \le 4$
- $-2 < x \le 4$
- $-2 < x < 4$
- $x \le 4$

$\sqrt{2x + 7} = x + 2$ gives:

- $x = 1$
- $x = 1, -3$
- $x = -3$
- none

$\sqrt{x + 5} - \sqrt{x} = 1$ gives:

- $x = 4$
- $x = 5$
- $x = 9$
- $x = 1$

$x - 6\sqrt{x} + 8 = 0$ gives:

- $x = 2, 4$
- $x = 4, 16$
- $x = 16$
- $x = 8$

$\sqrt{x} + \sqrt{y} = 6, x + y = 20$ gives:

- $(4,16)$ only
- $(4,16), (16,4)$
- $(2,4)$
- none

$\frac{2}{x-1} \ge 1$ gives:

- $x \ge 3$
- $1 < x \le 3$
- $x \le 3$
- $x > 1$

Domain of $\frac{\sqrt{x-2}}{x^2 - 25}$:

- $x \ge 2$
- $x \ge 2, x \ne 5$
- $x \ne \pm 5$
- $x > 2$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Equivalent transformation	Teng kuchli almashtirish	Равносильное преобразование
Non-equivalent step	Teng kuchli bo‘lmagan qadam	Неравносильный переход
Extraneous root	Chet ildiz	Посторонний корень
Sign chart	Ishoralar jadvali	Таблица знаков
Target	Maqsad	Цель
Self-assessment	O‘z-o‘zini baholash	Самооценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first line of any solution this quarter states:

- the answer
- what the unknown may be
- the method
- the check

Squaring is:

- always equivalent
- equivalent only when both sides are ≥ 0
- never allowed
- reversible

Quarter III begins with:

- trigonometry
- the exponential function
- probability
- vectors

A logarithm will demand:

- argument ≥ 0
- argument > 0
- base $= 10$
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $\sqrt{x} = 6$

ANSWER $x = 36$

2. Domain of $\frac{1}{x - 4}$

ANSWER $x \neq 4$

3. Solve $\frac{3}{x} = 1$

ANSWER $x = 3$

4. Solve $\frac{x - 4}{x + 2} \leq 0$

ANSWER $-2 < x \leq 4$

5. Solve $x + y = 6, xy = 8$

ANSWER $(2,4), (4,2)$

6. Domain of $\sqrt{x - 2}$

ANSWER $x \geq 2$

7. Solve $\sqrt{x + 1} = 3$

ANSWER $x = 8$

Medium · the standard question

7 PROBLEMS

1. Solve $\sqrt{2x + 7} = x + 2$

ANSWER $x = 1$

2. Solve $\sqrt{x + 5} - \sqrt{x} = 1$

ANSWER $x = 4$

3. Solve $x - 6\sqrt{x} + 8 = 0$

ANSWER $x = 4, 16$

4. Solve $\sqrt{x} + \sqrt{y} = 6, x + y = 20$

ANSWER $(4,16), (16,4)$

5. Solve $\frac{2}{x - 1} \geq 1$

ANSWER $1 < x \leq 3$

6. Domain of $\frac{\sqrt{x - 2}}{x^2 - 25}$

ANSWER $x \geq 2, x \neq 5$

7. Solve $\frac{x}{x + 3} = \frac{2}{x}$

ANSWER $x = 6, -1$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\sqrt{3x + 4} - \sqrt{x + 1} = 1$

ANSWER $x = 0$ or $x = 8$

2. Solve $\frac{x^2 - 1}{x^2 - 4} \le 0$

ANSWER $-2 < x \le -1$ or $1 \le x < 2$

3. Solve $\sqrt{x} + \sqrt{y} = 5, xy = 36$

ANSWER $(4,9), (9,4)$

4. Domain of $\sqrt{\frac{x - 3}{x + 2}}$

ANSWER $x < -2$ or $x \ge 3$

5. Solve $2x + \sqrt{x} - 1 = 0$

ANSWER $x = 0.25$

6. Find k so $\sqrt{x} = x - k$ has exactly two solutions

ANSWER $0 < k < 0.25$

7. Solve $\frac{1}{\sqrt{x}} + \sqrt{x} = 2.5$

ANSWER $x = 4$ or $x = 0.25$

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. Solve $\sqrt{4x + 5} = x + 2$ and $\sqrt{x + 3} + \sqrt{x} = 3$.

4. Write your target for Quarter III in one checkable sentence, and date it.